OS3-1

Quantification of bat offshore activity using passive acoustic monitoring above the North Sea

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We will present the results of one of the largest European pre-construction investigations of bat activity. The first year of the study included passive acoustic monitoring (PAM) with 22 buoy stations, 10 wind turbine stations, 11 land-based stations along the coast and recordings on a survey vessel in an area of the North Sea where multiple offshore wind parks are planned. To detect bat activity in the recordings we trained a convolutional neural network and showed that this method outperforms commercial software. Bats were recorded on 11 out of 22 offshore buoy stations, four out of 10 wind turbine deployments, and one out of 45 nights during vessel surveys. Most recordings were of Nathusius' pipistrelle (Pipistrellus nathusii), a known long-distance migrant, from mid-August to mid-September 2023, coinciding with temperatures > 15°C and wind speeds < 8 m/s. Nearly all bat passes were within 40 km of the coast, except for one record of two bats 80 km offshore. These patterns align well with the assumed general migration pattern across Europe for this species. We also recorded multiple buzz sequences suggesting foraging behaviour. On land, significant bat activity was recorded along the western Jutland coastline, with peaks in late August and September. We show that PAM of bat activity in locations far offshore is feasible, and can be used to quantify activity patterns as well as qualitatively assess what behaviours the bats are engaged in.

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OS3-2

Multi-task deep learning-based Anuran call classification and sound source separation

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Anuran calls are important for understanding the ecosystems in paddy environments and detecting invasive species therein. Passive Acoustic Monitoring (PAM) has emerged as a powerful, non-invasive method for continuously recording these vocalizations in natural habitats, enabling spatiotemporal tracking of target species. However, its effectiveness is often hindered by the overlapping vocalizations from multiple anuran species, which significantly reduce classification accuracy at the species level. In this study, we propose a multi-task deep learning model for sound source separation and classification for the East Japanese Tree Frog (Dryophytes leopardus) and the Black-Spotted Pond Frog (Pelophylax nigromaculatus) in Iwamizawa, Hokkaido, Japan. The multi-task deep learning model consists of a sound source separator for extracting target anuran calls from the recording data and a species classification module using feature maps of the separator. Spectrograms representing the time-frequency patterns of the anuran calls are used as input. The model was constructed to perform separation and classification in a step-by-step manner, aiming to maximize the SI-SDR loss as a separation performance measure and minimize cross-entropy loss as a classification performance measure. This multi-task approach is expected to improve model performance in a multi-objective manner, which can contribute to a deeper understanding of the breeding phenology monitoring of invasive alien species using the acoustic data from the real environment. In the future, we aim to develop a passive acoustic monitoring technique in which sound source separation, classification, and localization are integrated using spatiotemporal information derived from multi-channel microphone and array data.

OS3-3

Bugs in the system: Passive acoustic monitoring of tropical rainforest insect communities

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Insects play a crucial role in the terrestrial ecosystems, but are often labor intensive and difficult to census, particularly in habitats like the rainforest canopy. Despite urgent questions about the magnitude of change in insect populations and potential drivers of change, information on insect populations is often drawn from a relatively small number of locations and taxonomic groups. Passive acoustic monitoring (PAM) can provide detailed information on the spatial and temporal distribution of soniferous insects. When combined with machine learning approaches for automating event detection at multiple sites and multiple years, PAM can provide exceptionally detailed information about the dynamics of the calling insect community. We present a case study of multi-year insect acoustic monitoring on Barro Colorado Island in Panamá, with a particular focus on katydids and other soniferous insects. This monitoring integrates deployments in the forest canopy, as well as intensive deployment in wellcharacterized sites in the forest understory. We implemented a bespoke neural network model, supported by custom-engineered data-processing, to cope with the wide range of durations and bandwidths of the sounds of many of the focal species and conducted detailed evaluation of performance. We used this model to evaluate the spatial, temporal, and seasonal detection patterns of rainforest insects, test for impacts of moonlight, evaluate effects of habitat structure, and test for associations between the species diversity and acoustic detection rates of insects and birds. In changing forests, consistent approaches for insect sampling will be key to understanding insect ecology and generating interpretable and actionable data.

OS3-4

Underwater biophony of the Brazilian equatorial margin

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The Brazilian Equatorial Margin has been the subject of discussions regarding the exploration of its petroleum potential, and its environmental impacts in a region that has so far been minimally affected by human activities. Ongoing seismic operations and planned oil production result in shifts in the local soundscape dynamics. Such changes in the acoustic scenario may impact resident biodiversity as well as the behavioral and migratory patterns of non-resident animals, such as cetaceans, which rely on sound for communication, reproduction, navigation and feeding. We proposed a Passive Acoustic Monitoring study to characterize and identify changes on the local biodiversity. Six instrumented acoustic moorings were distributed over the 50 m and 1,000 m depth isobaths. Preliminary manual browsing of six months of recordings enabled the construction of a catalogue of 144 sonotypes. The majority (79.9 %) of sonotypes are from biological sources (biophony), 4.2% are from anthropic origin (anthropophony), and the remaining 15.9% sounds could not be classified into any category with certainty. The biophony was further categorized into sounds from: mysticetes (true whales; 27.8% of sonotypes), odontocetes (dolphins and toothed whales; 19.4%), fish (26.4%), and invertebrates (mostly crustaceans; 5.6%). Biophony varied over time, showing an abrupt increase in the acoustic presence of whales, from July 2nd onwards, suggesting the arrival of migratory whales in the region, especially humpback and Antarctic minke whales. It was also possible to notice the constant local presence of fish (mainly from the Sciaenidae family) in shallower waters and of dolphins in deeper areas.

OS3-5

Listening to the Amazonian freshwater giant: using air-breathing sounds to understand pirarucu (*Arapaima gigas*) dynamics in remote floodplain lakes

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This talk addresses key limitations inherent to soundscape-based classification, including imbalanced datasets with long-tailed distributions across both species and recording locations, weakly labeled or noisy training data and acoustic variability introduced by rare vocalizations, dialects and overlapping calls in rich soundscapes such as the dawn chorus. Additional complications arise from discrepancies and domain shifts between training conditions and real-world deployment scenarios.

OS3-6

Can you hear me now? The limits of individual yellowhammer song recognition

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Individual Acoustic Monitoring (IAM), based on vocal cue analysis, offers a non-invasive alternative to traditional mark-recapture methods. Combined with passive acoustic monitoring (PAM), it could provide new insights into fundamental ecological questions. However, few studies have examined how identity cues degrade over distance and across environments — a crucial factor in refining IAM and PAM approaches. In this study, we investigated how distance affects the acoustic structure of Yellowhammer (Emberiza citrinella) songs, aiming to determine the maximum distance at which individuals can still be reliably detected and assigned to a particular individual. We played back songs from ten males (20 songs per individual, representing their entire repertoire) in two environments: forest and grassland. We re-recorded the songs using Audiomoth devices along a 200 m transect (6.5 m to 200.0 m). We estimated the signal-to-noise-ratio (SNR) to describe sound degradation. Songs were assigned to a particular individual, based on the catalogue of song types and visual inspection of spectrograms, by two observers. Inter-observer agreement was evaluated using Cohen's kappa coefficient. We used a generalized linear mixed model (GLMM) to analyse the influence of habitat, distance, and SNR on ID assignments. Our results suggest that individual recognition remains possible at distances of up to approximately 100 metres, which is consistent with the typical size of Yellowhammer territories. We further found some differences in song transmission between the two habitats. These findings emphasise the importance of environmental context in IAM studies and provide a framework for future applications of PAM on individually identified species.

OS3-7

Narwhal repeated interaction with hydro-acoustic moorings as a matter of concern

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The melting of sea ice in the Arctic is expected to result in a greater presence of humans and infrastructure in the region. However, little is known about the likelihood of Arctic-endemic animals becoming entangled or interacting with human-made objects. We applied passive acoustic monitoring to study narwhals in Inglefield Bredning, Greenland. This method is known as a non-intrusive and transformative tool for ecology and has been increasingly used for conservation and biodiversity monitoring. However, the present study identified a high level of curiosity in narwhals with respect to deep-water scientific moorings (260–400 m) and explained recent cases of lethal entanglements in narwhals. Using acoustic data from different locations and years together with stomach content analysis, it is shown that foraging narwhals engaged in repeated hits on seafloor moorings (on average 11 times-per-day), presumably out of playful curiosity, to exfoliate or due to confusion with food items. It is a behavior previously unknown for unconfined odontocetes. These results imply that oceanographic monitoring might alter the natural behavior of narwhals and pose a risk to their well-being, which should be further investigated and accounted for in the design. Moreover, physical shocks to some popular acoustic releasers at any depth can result in a considerably higher pressure on the case, and water may enter the receiver and damage the internal electronics. Our findings reveal the intrusive nature of a key scientific method and should inform policy-making concerned with the management/conservation of narwhal habitats.

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Edge-based machine learning for real-time bittern population monitoring and individual identification.

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We present a novel approach to real-time monitoring of Eurasian Bittern populations using edge-based machine learning on solar-powered devices. Our system leverages Raspberry Pi-based nodes to record and process bird calls in real-time, enabling efficient and accurate population monitoring. European Bitterns are secretive (Cryptospecies), territorial and have Schedule One conservation status. We present a three-stage framework we have developed that has been designed to address the unique challenges of detecting individual male Bitterns. Stage 1: Is that a Bittern? Initially, our system determines if a call belongs to a Bittern. This is achieved through machine learning-based audio classification using a pre-trained model deployed on Raspberry Pi Zero 2's. Stage 2: "Where is he calling from?" Once a bittern is detected, the system triangulates the caller's location by calculating the bearing from each monitoring node, which allows for accurate population tracking. Stage 3: "Have we heard this male before?" The final stage of our framework focuses on individual identification. Using a separate machine learning model, trained on audio deltas obtained from captured calls, such as spectral characteristics, time-frequency representations, or even raw waveforms. This enables the system to recognize and distinguish between individual males. By combining edge-based processing with real-time data analysis, our system offers a promising solution for real-time efficient low-cost and accurate population monitoring of Eurasian Bitterns.

OS3-9

Acoustically investigating distributions and behavior of overwintering northeast Pacific blue whales using ocean bottom seismometer networks

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Northeast Pacific (NEP) blue whales are specialist krill feeders that have historically been described as capital breeders, feeding off the coast of Southern California in the summer and breeding in the Eastern Tropical Pacific in the sinter. However, this stereotypical pattern is challenged by the presence of NEP blue whale song recorded on passive acoustic monitoring (PAM) instruments in the North Pacific throughout the winter. The goal of our study is to investigate the behavior of whales that skip southward migration and overwinter off the coast of the Pacific Northwest through analyzing song produced by males recorded by ocean bottom seismometers (OBSs). OBSs are opportunistic PAM instruments; they are designed to record ground motion from earthquakes, but also record low-frequency baleen whale calls. Our dataset includes ~70 OBSs from two coincident seismic experiments deployed from Northern California to British Columbia from 2013-2014. We acoustically localize B-calls that arrive on multiple OBSs using time-difference-of-arrival methods. We then associate nearby localizations into tracks that describe the movement of individual whales. Spatial variations in movement behavior and day-night song distributions identify the southern area of our study region as a migration corridor, and the northern area of our study region as an overwintering and foraging ground. We also link song phrase characteristics with behavior by associating call metrics with whale movement. We find that song phrases include more B-notes later in the year, and when whales are undertaking migrational movement defined by travelling at fast swimming speeds and at N-S azimuths.